

Poverty

Poverty

If somebody is earning “a dollar a day” he is living on the poverty line. If earning less than one dollar, he is poor.

Absolute poverty

The situation of being unable or only barely able to meet the subsistence essentials of food, clothing and shelter.

Extent of poverty

The number of people who are unable to command sufficient resources to satisfy basic needs. They are counted as the total number living below the poverty line (\$ 1.25 a day).

Absolute poverty index/Head count index

Absolute poverty is sometimes measured by the number or “Head count” H of those whose incomes fall below the absolute poverty line Y_p . When the head-count is taken as a fraction of the total population, N , we call it Headcount index

The proportion of a country's population living below the poverty line.

Headcount index = H/N .

Relative poverty

Total Poverty Gap (TPG)

It measures the total amount of income necessary to raise everyone, who is below the poverty line, up to that line. It is equal to the shaded **area between poverty line and annual income profile** of the country.

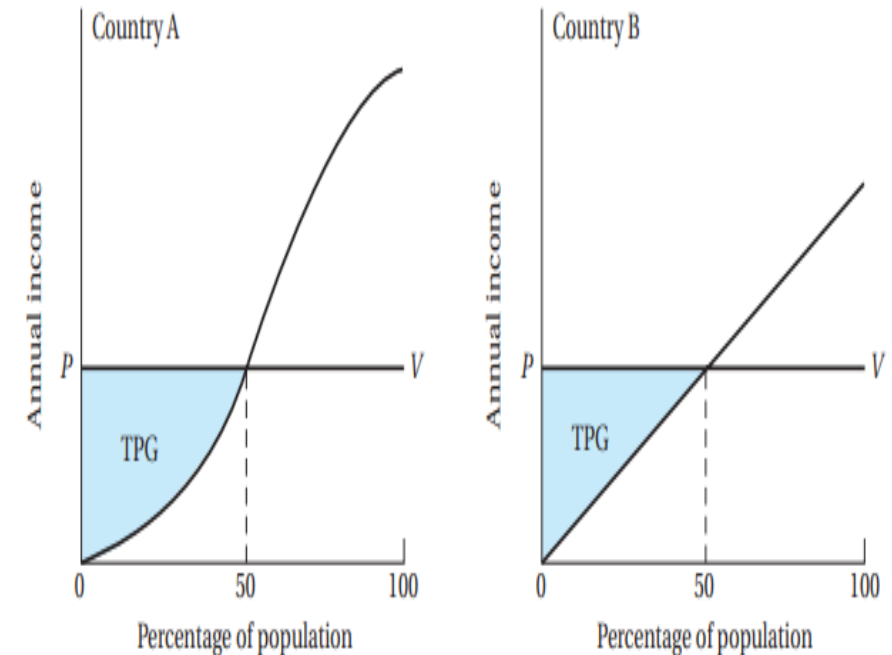
The extent to which the incomes of the poor lie below the poverty line...is found by adding up the amounts by which each poor person's income Y_i falls below the absolute poverty line Y_p .

$$TPG = \sum (Y_p - Y_i)$$

TPG can be thought of as the amount of money per day it would take to bring every poor person in an economy up to our defined minimum income standards.

Country A is relatively more poor than country B.

FIGURE 5.6 Measuring the Total Poverty Gap



(a) A relatively large poverty gap

(b) A relatively small poverty gap

Measuring inequality and poverty: size distribution

The personal or size distribution of income is the measure most commonly used by economists. It simply deals with individual persons or households and the total incomes they receive.

1. The way in which that income was received is not considered
2. Moreover, the locational (urban or rural) and occupational sources of the income (e.g., agriculture, manufacturing, commerce, services) are ignored

Economists and statisticians therefore like to arrange all individuals by ascending personal incomes and then divide the total population into distinct groups, or sizes.

A common method is to divide the population into successive quintiles (fifths) or deciles (tenths) according to ascending income levels and then determine what proportion of the total national income is received by each income group

A common measure of income inequality that can be derived from column 3 is the **ratio of the incomes received by the top 20% and bottom 40%** of the population. This ratio, sometimes called a **Kuznets ratio** after Nobel laureate Simon Kuznets, has often been used as a measure of the degree of inequality between high- and low-income groups in a country. In our example, this inequality ratio is equal to **51 divided by 14, or approximately 3.64**.

TABLE 5.1 Typical Size Distribution of Personal Income in a Developing Country by Income Shares—Quintiles and Deciles

Individuals	Personal Income (money units)	Share of Total Income (%)	
		Quintiles	Deciles
1	0.8		
2	1.0		1.8
3	1.4		
4	1.8	5	3.2
5	1.9		
6	2.0		3.9
7	2.4		
8	2.7	9	5.1
9	2.8		
10	3.0		5.8
11	3.4		
12	3.8	13	7.2
13	4.2		
14	4.8		9.0
15	5.9		
16	7.1	22	13.0
17	10.5		
18	12.0		22.5
19	13.5		
20	15.0	51	28.5
Total (national income)	100.0	100	100.0

Lorenz curve

A graph depicting the variance of the size distribution of income from perfect equality

The Lorenz curve shows the **actual quantitative relationship between the percentage of income recipients and the percentage of the total income** they did in fact receive during, say, a given year.

The numbers of income recipients are plotted on the horizontal axis, not in absolute terms but in cumulative percentages

The vertical axis shows the share of total income received by each percentage of population

both axes are of the same length

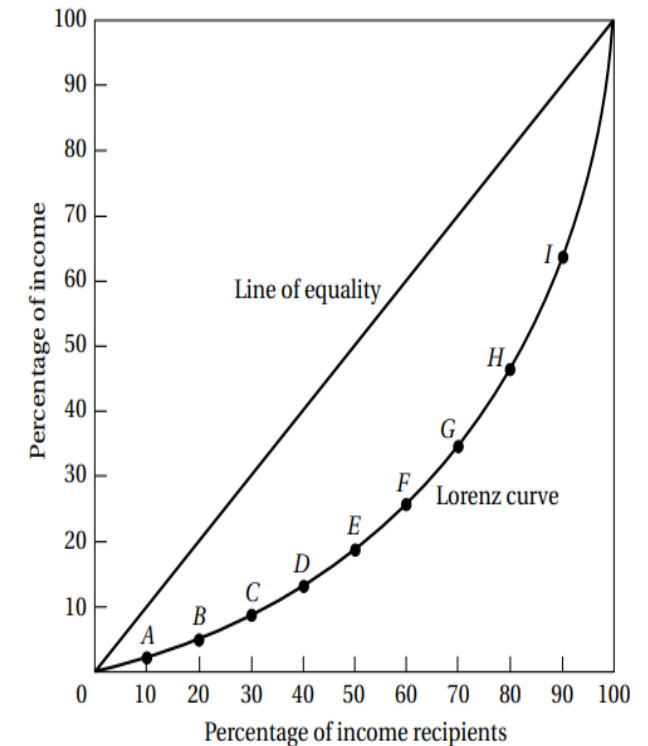
At every point on that diagonal, the percentage of income received is exactly equal to the percentage of income recipients—for example, the point halfway along the length of the diagonal represents 50% of the income being distributed to exactly 50% of the population

The diagonal line in Figure 5.1 is representative of “perfect equality” in size distribution of income.

The more the Lorenz curve away from the diagonal (line of perfect equality), the greater the degree of inequality represented

Because no country exhibits either perfect equality or perfect inequality in its distribution of income, the Lorenz curves for different countries will lie somewhere to the right of the diagonal

FIGURE 5.1 The Lorenz Curve



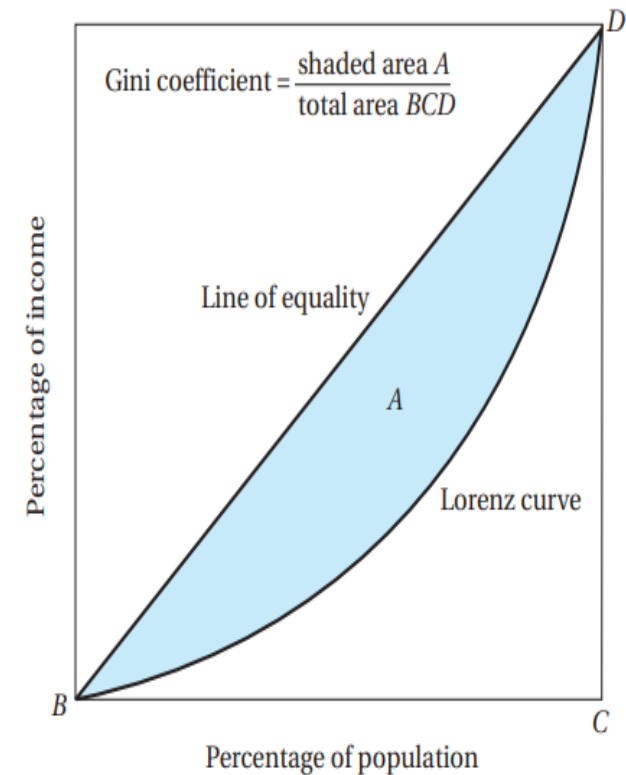
Ginni coefficient and aggregate measure of inequality

An aggregate numerical measure of income inequality ranging from 0 (perfect equality) to 1 (perfect inequality). It is measured graphically by dividing the area between the perfect equality line and the Lorenz curve by the total area lying to the right of the equality line in a Lorenz diagram. This is the ratio of the shaded area A to the total area of the triangle BCD. This ratio is known as the Gini concentration ratio or Gini coefficient, named after the Italian statistician who first formulated it in 1912.

The higher the value of the coefficient, the higher the inequality of income distribution; the lower it is, the more equal the distribution of income.

Highly unequal income distributions typically lies between 0.50 and 0.70, while for countries with relatively equal distributions, it is on the order of 0.20 to 0.35

FIGURE 5.3 Estimating the Gini Coefficient



Functional distribution

The second common measure of income distribution used by economists, the functional or factor share distribution of income, attempts to explain the share of total national income **that each of the factors of production (land, labor, and capital) receives.**

It attempts to explain the income of a factor of production by the contribution that this factor makes to production.

Supply and demand curves are assumed to determine the **unit prices** of each productive factor. When these unit prices are multiplied by quantities employed on the assumption of efficient (minimum-cost) factor utilization, we get a measure of the total payment to each factor. For example, the supply of and demand for labor are assumed to determine its market wage. When this wage is then multiplied by the total level of employment, we get a measure of total wage payments, also sometimes called the total wage bill

Functional distribution

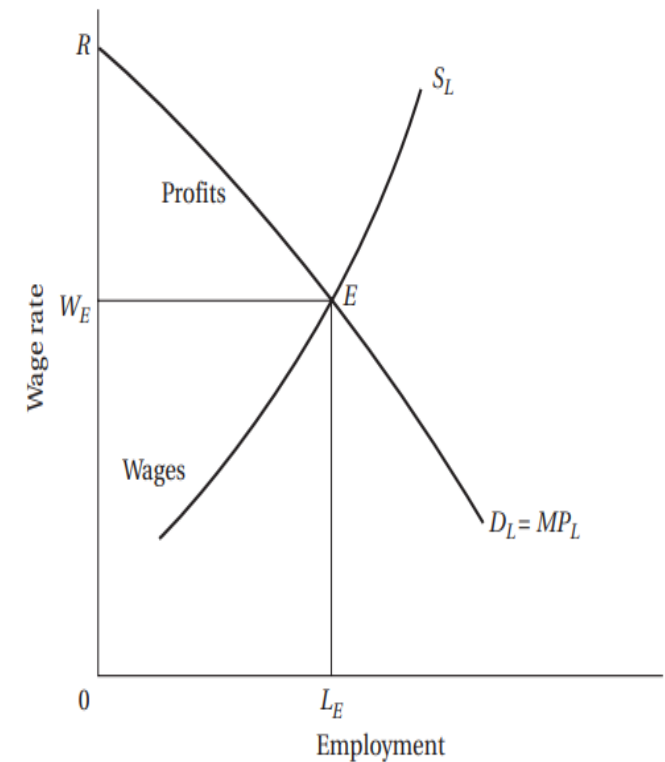
We assume that there are only two factors of production: capital, which is a fixed factor, and labor, which is the only variable factor.

Under competitive market assumptions, the demand for labor will be determined by labor's marginal product. Such a negatively sloped labor demand curve is shown by line D_L

A traditional neoclassical upward-sloping labor supply curve is S_L . The equilibrium wage will be equal to W_E and the equilibrium level of employment will be L_E . Total national output will be represented by the area $0REL_E$. This national income will be distributed in two shares: $0W_EEL_E$ going to workers in the form of wages and W_ERE remaining as capitalist profits (the return to owners of capital)

Every factor gets paid only in accordance with what it contributes to national output, no more and no less.

FIGURE 5.5 Functional Income Distribution in a Market Economy: An Illustration



Poverty, inequality and welfare

Social welfare depends: positively on the level of **income per capita**, negatively on **poverty**, negatively on the level of **inequality**

If our top priority is the alleviation of absolute poverty, why should relative inequality be a concern?

Why should we be concerned with inequality among those above the poverty line?

There are three major answers to this question.

1. First, extreme income inequality leads to economic inefficiency

- lack of collateral.

- overall rate of saving in the economy tends to be lower

2. Extreme income disparities undermine social stability and solidarity.

3. Finally, extreme inequality is generally viewed as unfair

- welfare = $W(Y, I, P)$

- Y is income per capita and enters positively

- I is inequality and enters negatively

- P is absolute poverty and also enters negatively

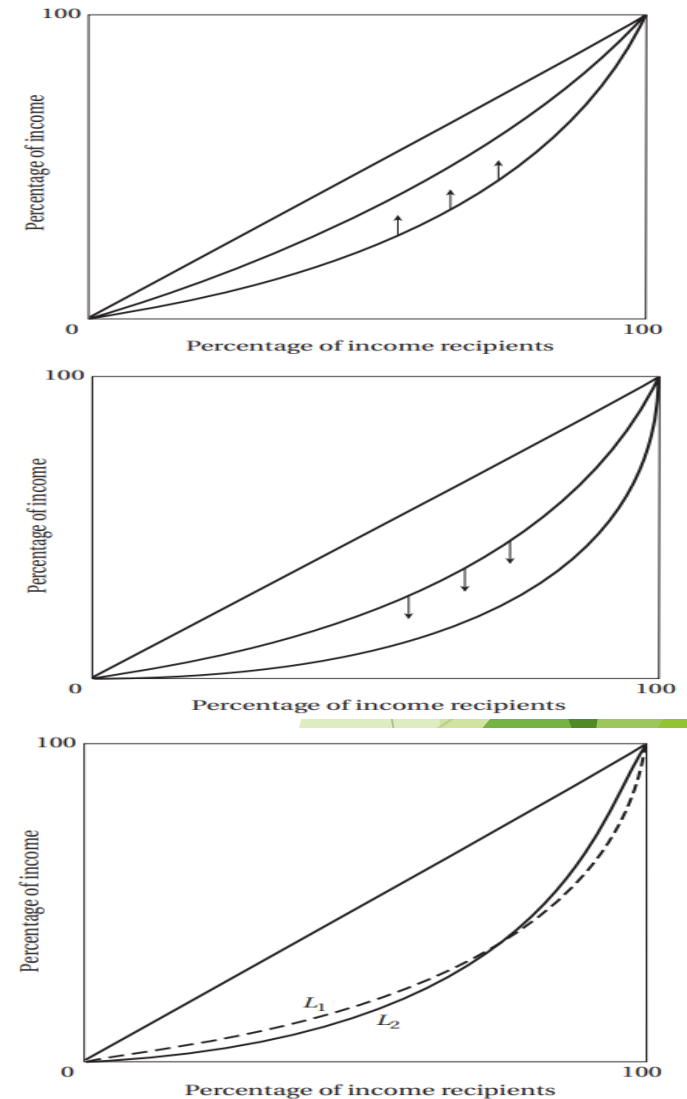
Dualistic Development and Shifting Lorenz Curves: Some Stylized Typologies

The **traditional-sector enrichment growth** typology, in which all of the benefits of growth are divided among traditional-sector workers, with **little or no growth occurring in the modern sector**.

a more **equal distribution of income, and less poverty**. causes the Lorenz curve to shift uniformly upward and closer toward the

The **modern-sector enrichment** growth typology, in which the economy grows but such growth is limited to a fixed number of people in the modern sector, with both the **numbers of workers and their wages held constant in the traditional sector**. Results in higher incomes, **a less equal distribution of income, and no change in poverty**. Causes the Lorenz curve to shift downward and farther from the line of equality.

The **modern-sector enlargement** growth typology, in which the two-sector economy develops by enlarging the size of its modern sector while maintaining **constant wages in both sectors**. **Absolute incomes rise and absolute poverty is reduced**, but the Lorenz curve will always cross, indicating that we cannot make any unambiguous statement about changes in relative inequality



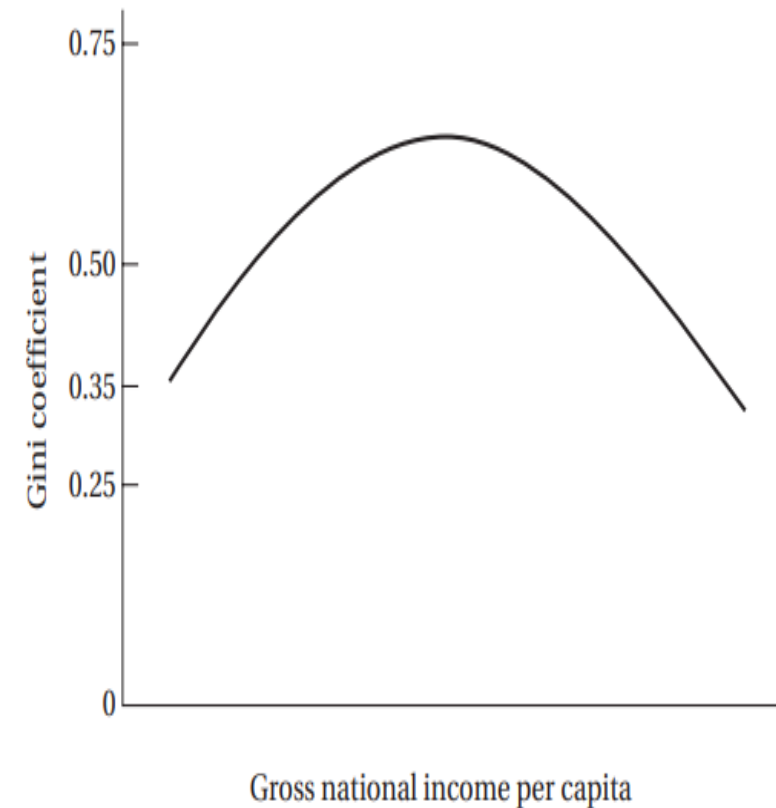
Kuznets's Inverted-U Hypothesis

A graph reflecting the **relationship between a country's income per capita and its equality of income distribution**

Simon Kuznets suggested that in the early stages of economic growth, the distribution of income will tend to worsen; only at later stages it will improve. This observation came to be characterized by the “inverted-U” Kuznets curve because time-series plot of changes in the distribution of income— as measured by the Ginni coefficient—seemed, when per capita GNI expanded, to trace out an inverted U-shaped curve

1. Early growth may be concentrated in the modern industrial sector, where employment is limited but wages and productivity are high
2. Returns to education may first rise as the emerging modern sector demands skills and then fall as the supply of educated workers increases and the supply of unskilled workers falls

FIGURE 5.10 The “Inverted-U” Kuznets Curve



Economic characteristic of High poverty groups

Rural poverty

The poor are disproportionately located **in rural areas**, that they are primarily engaged in agricultural and associated activities, that they are **more likely to be women and children** than adult males, and that they are often concentrated **among minority ethnic groups** and indigenous peoples.

Govt expenditures are urban biased.

Women and poverty

Women and children experience the harshest deprivation. They are more likely to be poor and malnourished and less likely to receive medical services, clean water, sanitation and other benefits.

The prevalence of female-headed household, the lower earning capacity of women, and their limited control over their spouse's income.

Women have less access to education, formal sector employment, social security, and Govt. employment programs.

A disproportionate number of ultra poor live in household headed by women.

Distribution of income within the household may be quite unequal.

Ethnic minorities, indigenous populations and poverty

Poverty falls heavily on minority ethnic groups and indigenous populations.

40% of world's states have more than five ethnic populations

Poverty is even more serious for indigenous peoples.

Poor countries

The poor come from poor countries.

Poverty negatively related to per capita income.

Poverty of opportunity index POI

Mehboob Ul Haq Human Development Center, Islamabad has introduced this index.

Poverty is not on the basis of just income, rather it is poverty on the basis of opportunities.

Income based poverty is just one deprivation, other poverty is deprivation of opportunities.

Poverty of opportunity is the cause and poverty is the result.

Strategies to eliminate poverty must aim at the abolition of poverty of opportunities for the poor.

Following arguments are given in this respect

1. The growth level will have to be more equitable i.e. easy access for the poor to the education and health etc.
2. Major share of public resources must be diverted to rural areas, as the rural poverty is higher than urban poverty.
3. Decentralized planning should be promoted.
4. Those growth models should be followed which are concerned with human development. They will increase human capabilities and opportunities.

